Painting with Darts: A Language-Model Orchestrator over a Fine-Tuned $\pi_{0.7}$ Policy on an SO-101 Arm

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Act-athon 2026 (Actor Labs × Physical Intelligence) · 3rd place

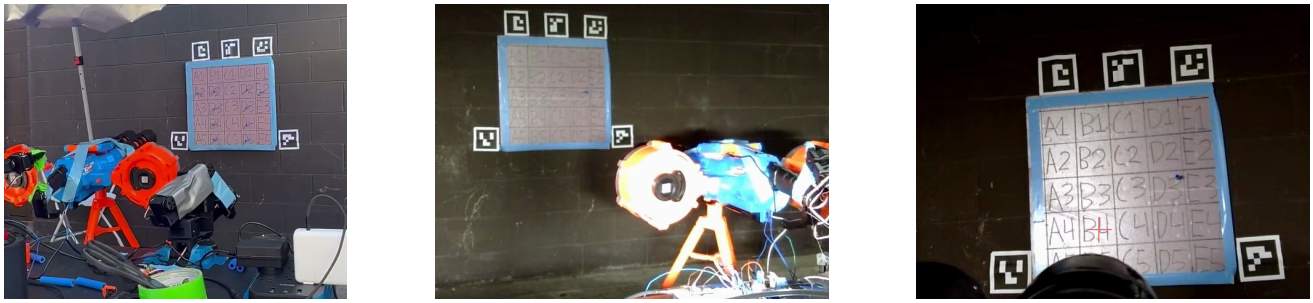


Figure 1: Left: the rig. An SO-101 arm drives the handle of a motorized dart blaster on a tripod, aimed at a 5×5 foam board. Middle and right: the two policy cameras at the moment of a `shoot B4` demonstration. The rear camera sees the arm, the blaster and the board. The wrist camera rides on the blaster; the red crosshair marks where the barrel points.

Abstract

We present a two-level system that draws pictures on a 5×5 grid by firing foam darts from a motorized blaster held by an SO-101 arm. A language model acts as the *orchestrator*: given a goal such as "draw the PI logo" and photos of the board, Claude plans the order of cells, checks the board after every shot and re-issues shots that missed. A vision-language-action (VLA) model acts as the *policy*: $\pi_{0.7}$, fine-tuned on our teleoperation data, maps the instruction `shoot <cell>`, two camera views and the arm state to 50-step chunks of joint targets plus a trigger channel. We collected 116 leader–follower demonstrations (29,131 frames at 30 Hz, 16 minutes) covering 23 of the 25 cells, with varied board clutter and start poses. On held-out frames the best fine-tune reaches a validation error of 0.287, and all 24 probed checkpoints aim in column and row order when only the cell in the prompt changes. The system completed its live drawing demo. We release the dataset in LeRobot format under CC BY 4.0, with the code, wiring and training logs.

1 System

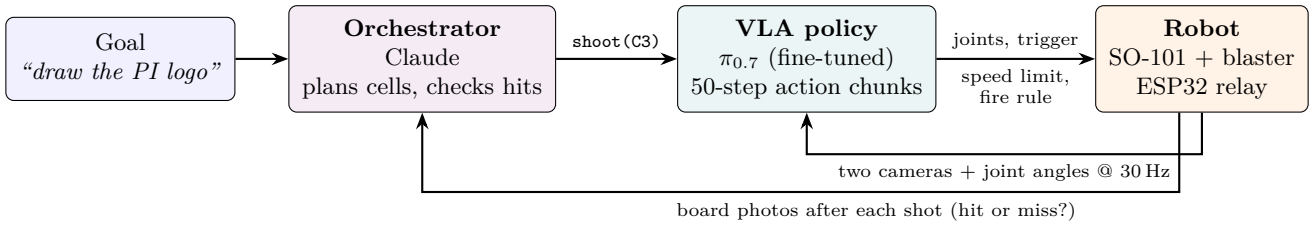


Figure 2: Two-level control. The orchestrator works at the level of shots (seconds). The VLA policy works at the level of joints (30 Hz).

Orchestrator. Claude receives the goal and two photos: the rear view, which shows which cells already hold darts, and the gun view with its fixed crosshair. It returns one tool call per turn (`shoot(cell)` or `finish`). After each shot it sees new photos, so a dart that bounced off or landed in the wrong cell is detected and the cell is shot again. It avoids the two cells that have no demonstrations.

VLA policy. $\pi_{0.7}$ receives both 640×480 frames, the five follower joint angles and the prompt `shoot <cell>`. It returns 50 future steps of absolute joint targets and a trigger value. We execute 25 steps per request, limit every joint to 3° per step, and fire only when the arm is still and the trigger exceeds 0.5 for three consecutive steps (or the remaining plan holds the aim and contains such a run).

Hardware. The arm's gripper is replaced by a wire loop through the blaster handle. The blaster's fire microswitch is replaced by a relay driven by an ESP32 (GPIO13), so firing is a single digital output. A rocker switch in the battery line is the master power: off gives dry shots with identical motion. Darts carry thumbtack tips so hits stay in the foam. Full wiring is in the repository.

2 Dataset

Operators aimed the follower with a leader arm and pressed a physical button to fire. A web app proposed the next cell, board clutter and start pose, overlaid the crosshair, and showed each take frame by frame for keep-or-delete review.

Episodes / frames	116 / 29,131 at 30 Hz (16.2 min); 142–505 steps each
Cells	23 of 25 (A5, E5 absent); task string `shoot <cell>`
Observation	rear + wrist RGB (480×640), follower joints (5, degrees)
Action	leader joints (5, degrees) + trigger (1 for 10 steps after the button)
Variation	23 live shots, 93 dry; board clutter light/heavy/late; start at rest or aimed at another cell
Format	LeRobot v3.0 (parquet + AV1), CC BY 4.0

Table 1: Dataset summary. huggingface.co/datasets/vineethsendilraj/pi07-dart-painting-so101

3 Training and evaluation

We fine-tuned $\pi_{0.7}$ with Physical Intelligence's hosted fine-tuning service (13 runs). Validation error was lowest after about 30–35 passes over the data; longer runs and a larger batch (512) overfit or did worse. The best run, `full-v2` (3,800 steps, about 33.5 passes), reached 0.287. In an offline prompt probe we fed the same rig frames and joint state with every `shoot <cell>` prompt; all 24 checkpoints produced aims ordered by column and row, so the policy follows the language instruction rather than replaying memorized routes.

4 Limitations

We did not measure a real-world hit rate; the evidence is offline validation, the prompt probe and the completed live demo. Each take records 2 s after the shot with `trigger=0`, so on-target frames mostly say "do not fire". Offline, the policy rarely produced a clean trigger pulse, which motivated the fire rule above. Coverage is uneven (2 demonstrations for row 1, A3 and E3), and the rig geometry is fixed, so the policy is not expected to transfer to a moved board or camera.

Code, wiring and training logs: github.com/VineethSendilraj/pi07-dart-painting-so101. Thanks to Actor Labs and Physical Intelligence for the event and API access.